

EE645 — Physical Models in Remote Sensing

Final Report: Discrete Ordinates Method

Vegetation Canopy Radiative Transfer
DOM_v3 — Bi-Lambertian Phase Function

Brian DeVoe
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Prof. Ranga B. Myneni

Energy Balance Table

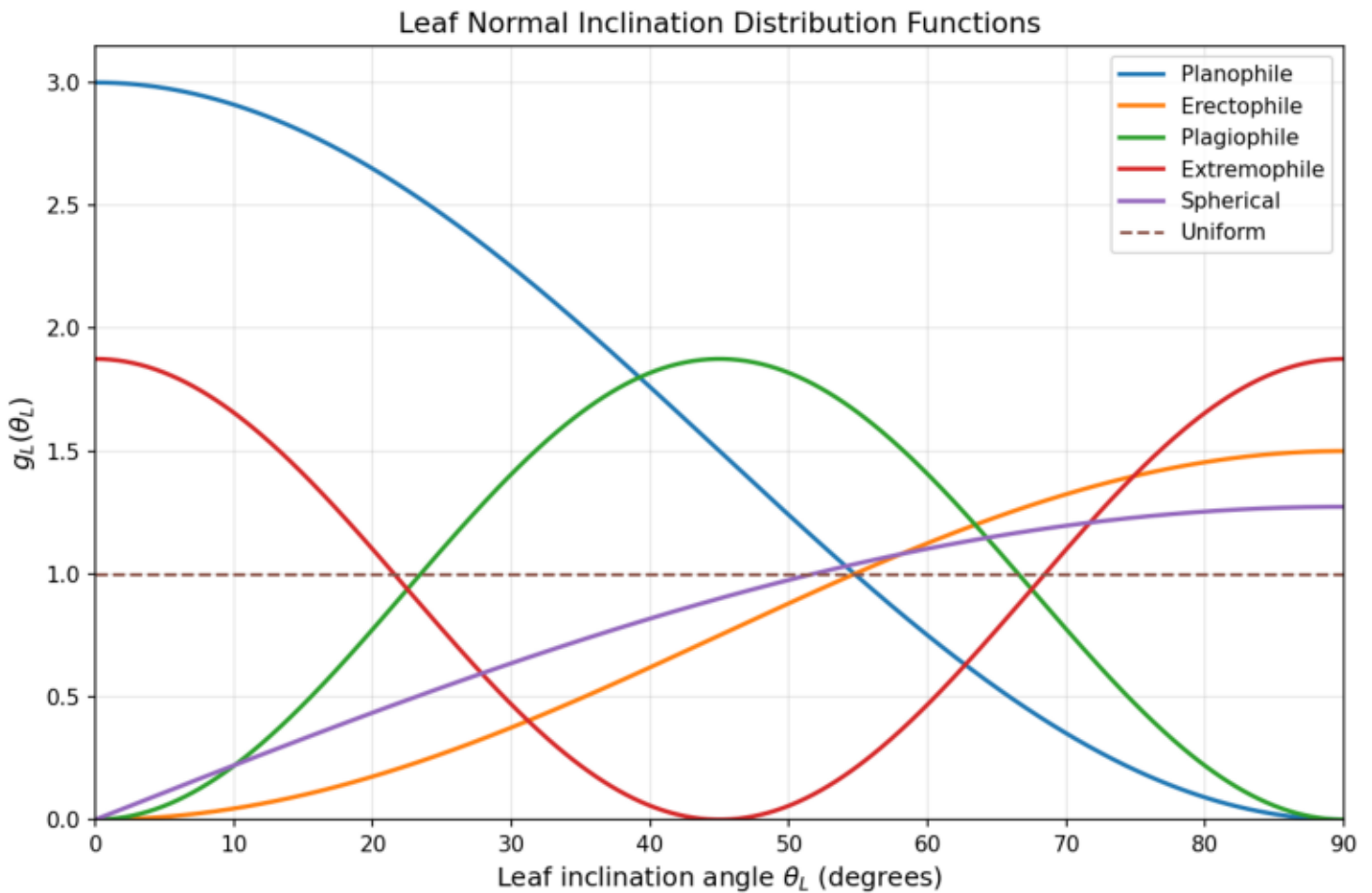
Energy Balance Table — DOM_v3
fdir=1.0 (pure direct beam → DHR), SZA=40°, N=16, M=16, K=50, uniform LAD (G=0.5)

Configuration	DHR	Transmittance	Canopy Abs.	Ground Abs.	Imbalance (%)
RED LAI=1.5	0.0318	0.3887	0.6185	0.3498	-0.013%
RED LAI=4.0	0.0223	0.0793	0.9065	0.0713	-0.020%
NIR LAI=1.5	0.3214	0.6359	0.1330	0.5405	+0.512%
NIR LAI=4.0	0.4177	0.3100	0.3089	0.2635	+0.987%

DHR = Directional-Hemispherical Reflectance (= BHR when fdir=1.0). Transmittance = $F_{\downarrow_ground} / F_{\downarrow_in}$. Imbalance = $F_{\downarrow_in} - (DHR + \text{Canopy Abs.} + \text{Ground Abs.})$.

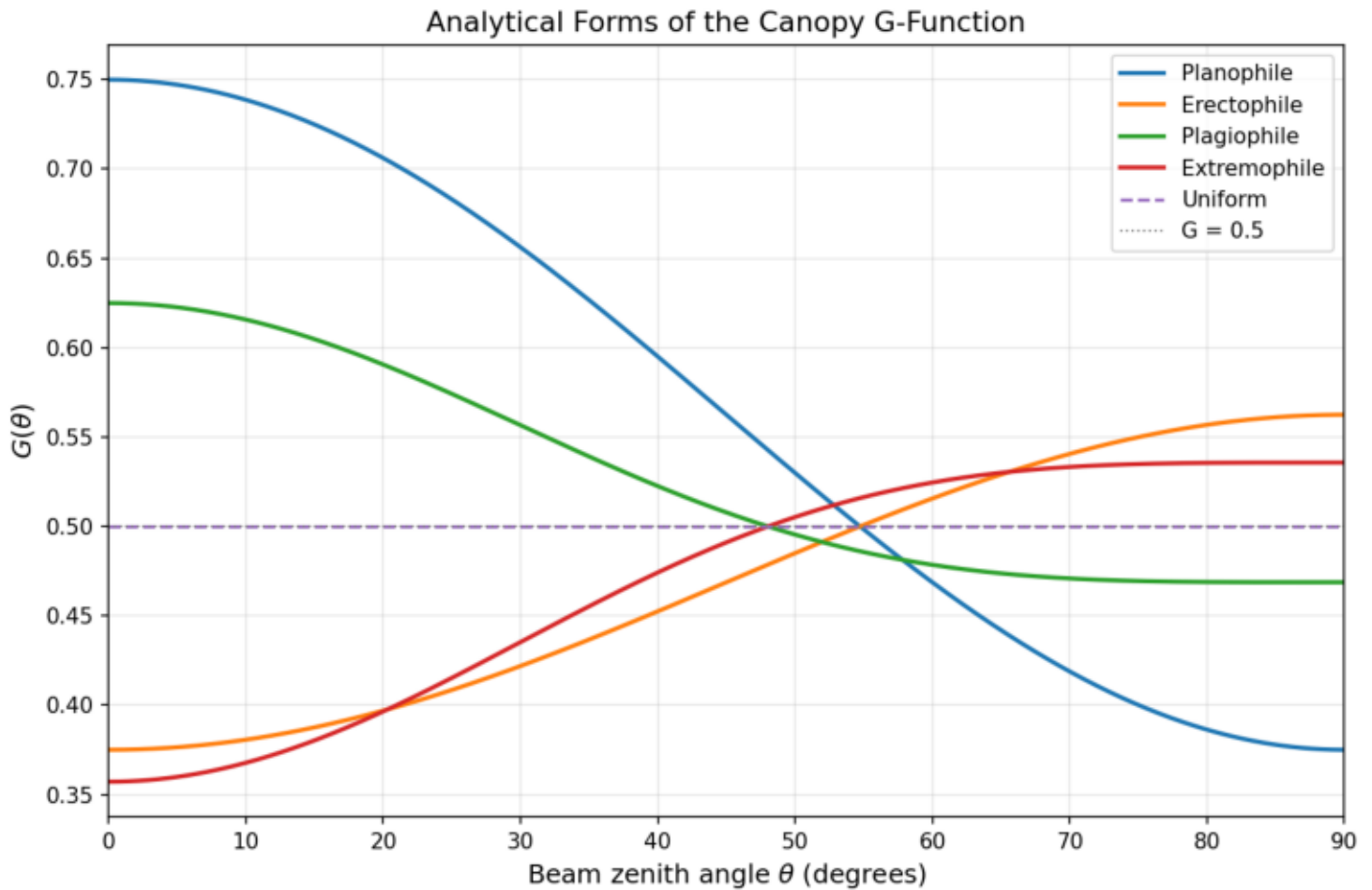
Canopy energy budget for RED and NIR bands at LAI=1.5 and 4.0, with fdir=1.0 (pure direct beam), SZA=40°. Reflectance equals the Directional-Hemispherical Reflectance (DHR) since the illumination is purely collimated. RED band: leaves absorb most incoming radiation (62-91%), yielding very low DHR (2-3%). NIR band: leaves scatter strongly, giving high DHR (32-42%) and lower canopy absorption (13-31%). Energy imbalance < 1% for all cases confirms solver convergence. Spectral parameters — RED: rhoL=0.06, tauL=0.04, rhoG=0.10; NIR: rhoL=0.45, tauL=0.45, rhoG=0.15.

Figure 1 — Leaf Angle Distributions



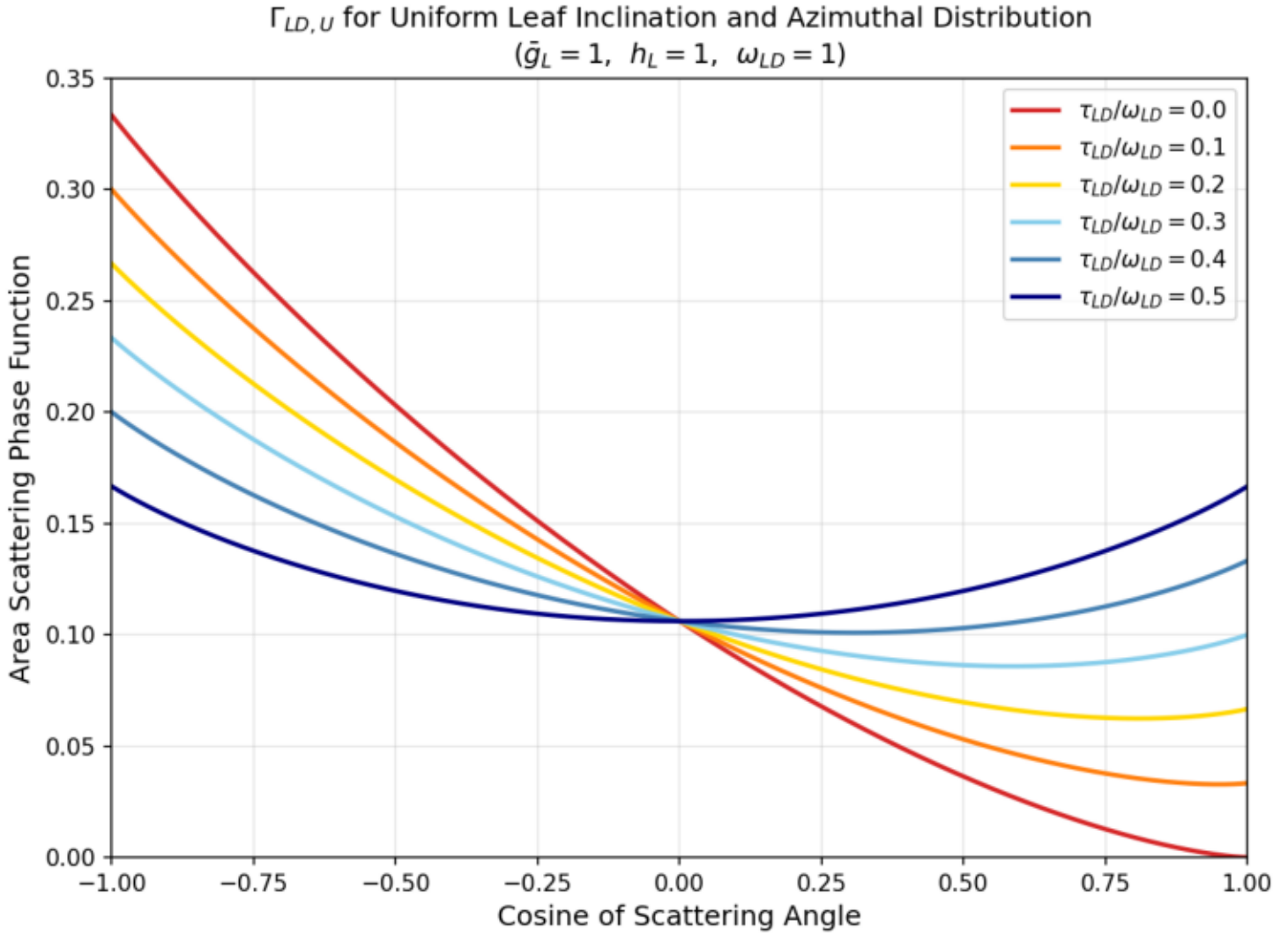
The six leaf inclination angle distribution functions $g(\theta_L)$ plotted vs. θ_L from 0 to 90 degrees: uniform, planophile, erectophile, plagiophile, extremophile, and spherical. Each distribution describes the statistical orientation of leaves. Planophile canopies have mostly horizontal leaves; erectophile have mostly vertical. The uniform (spherical) distribution gives $G=0.5$ for all solar/view directions and is used throughout this report.

Figure 2 — G-Function for Various Leaf Angle Distributions



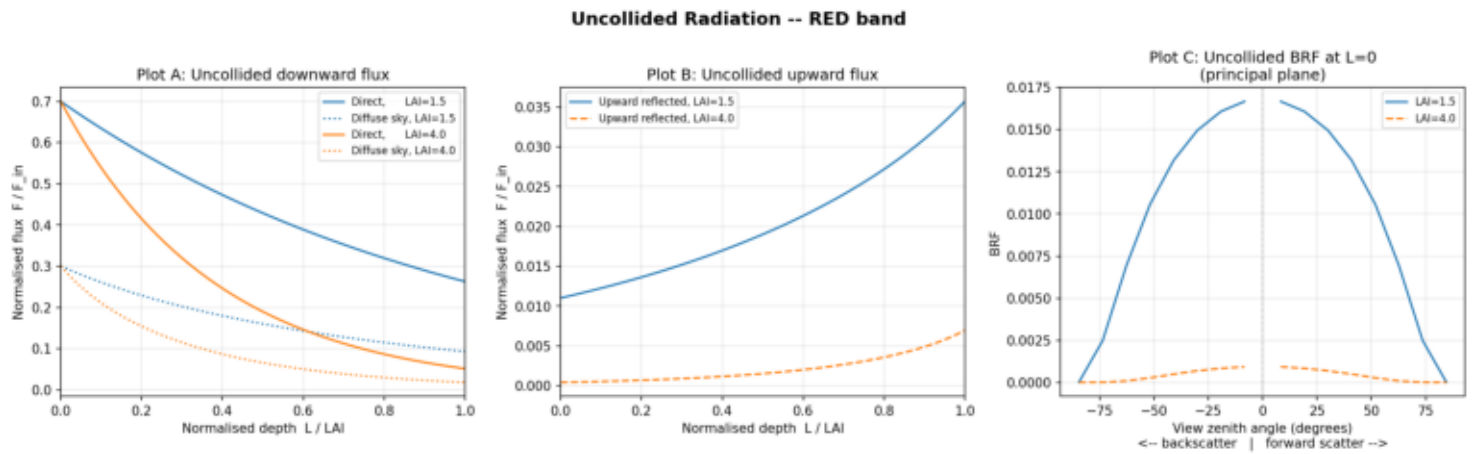
The Ross-Nilson G-function $G(\theta)$ vs. solar/view zenith angle for five LADs (uniform, planophile, erectophile, plagiophile, extremophile). $G(\theta)$ is the mean projected leaf area per unit leaf area in direction θ . All distributions satisfy the identity $(1/2\pi) * \int G \, d\Omega = 0.5$. The uniform LAD gives $G=0.5$ for all angles — this is the assumption used in DOM_v3.

Figure 3 — Volume Scattering Phase Function



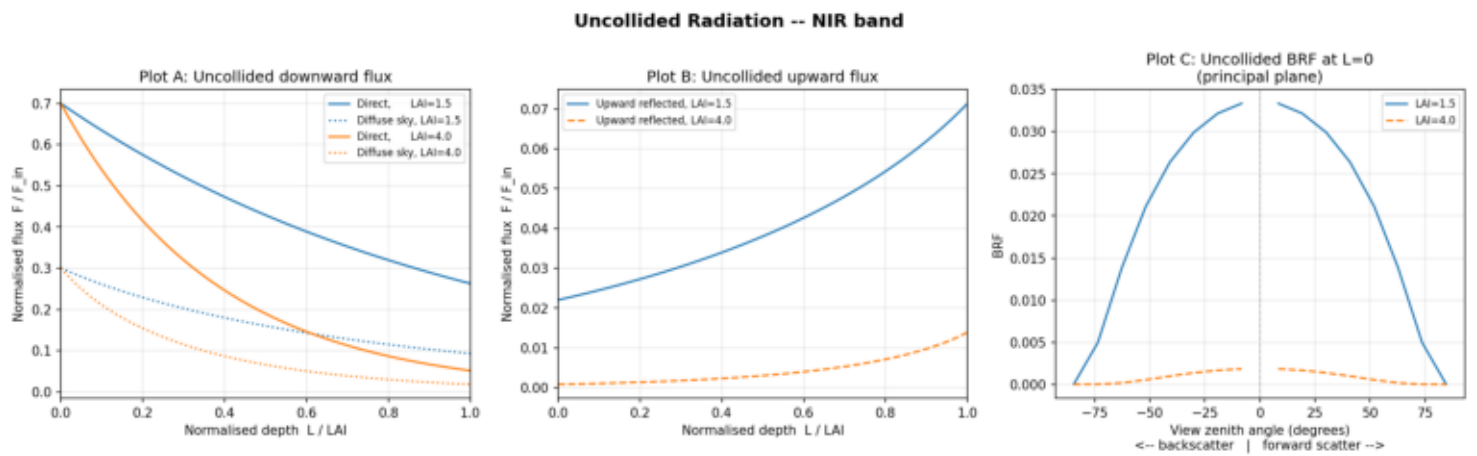
The bi-Lambertian area scattering phase function $\Gamma(\cos \beta)$ for uniform LAD, shown for several τ/ω ratios (0.0 to 0.5). The phase function governs how a leaf scatters incident radiation: the τ_L term drives forward scattering (transmission) and the ρ_L term drives backward scattering (reflection). The function is smooth with no singularity at $\beta=180^\circ$ (retro-illumination); the hot-spot effect (Ch4 Sec.9) is not modelled in DOM_v3.

Figure 4 — Uncollided Radiation: RED Band



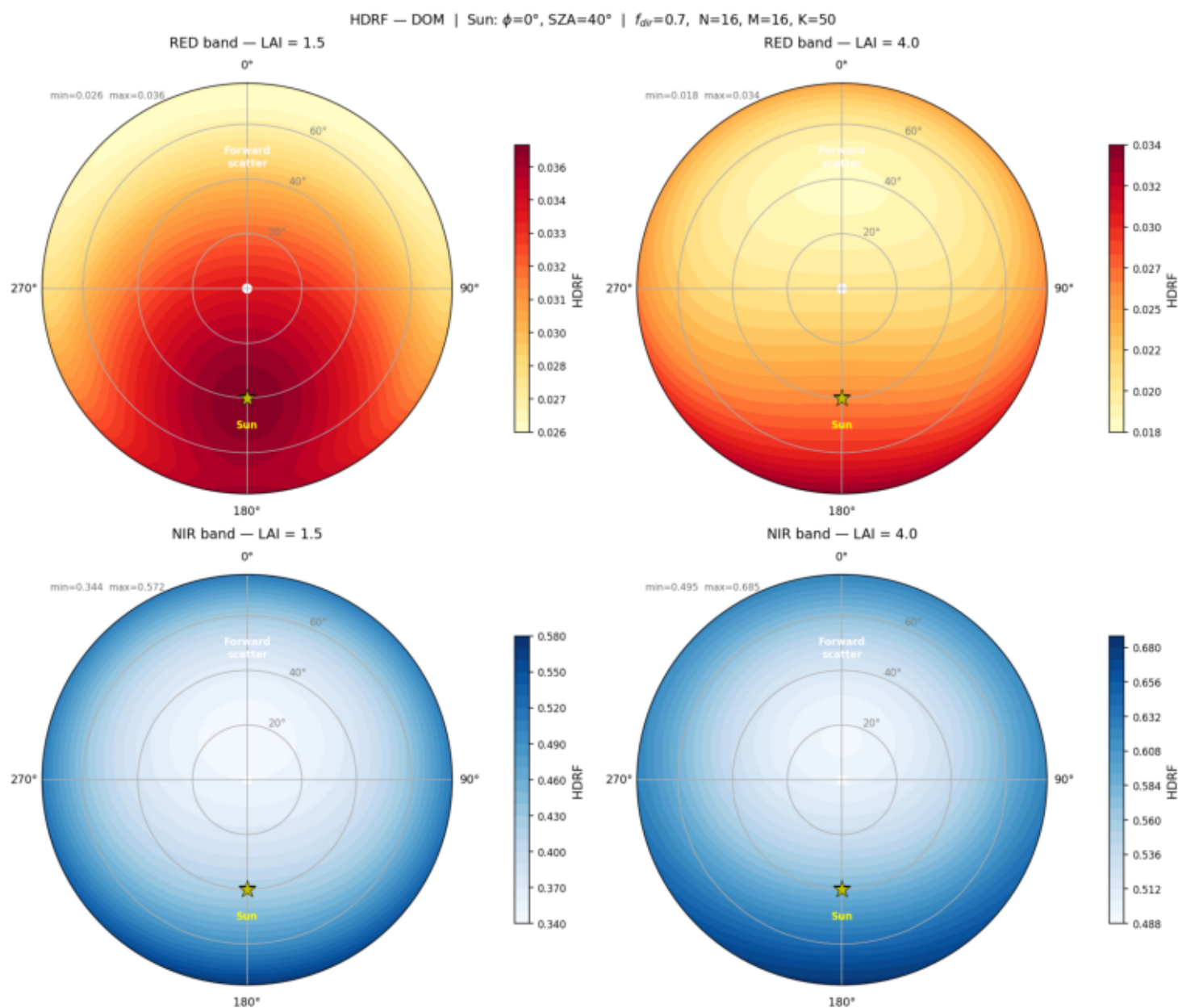
Three-panel uncollided DOM solution for the RED band ($\rho_L=0.06$, $\tau_L=0.04$, $\rho_G=0.10$). Plot A: normalised downward uncollided flux (direct solar + diffuse sky) vs. normalised canopy depth L/LAI — Beer-Lambert exponential decay with a steeper slope for larger LAI. Plot B: normalised upward ground-reflected uncollided flux — low values because strong leaf absorption limits what reaches the ground. Plot C: uncollided BRF at canopy top in the principal plane. Parameters: SZA=40 deg, $f_{dir}=0.70$, LAI=1.5 and 4.0, $N=M=16$, $K=50$.

Figure 5 — Uncollided Radiation: NIR Band



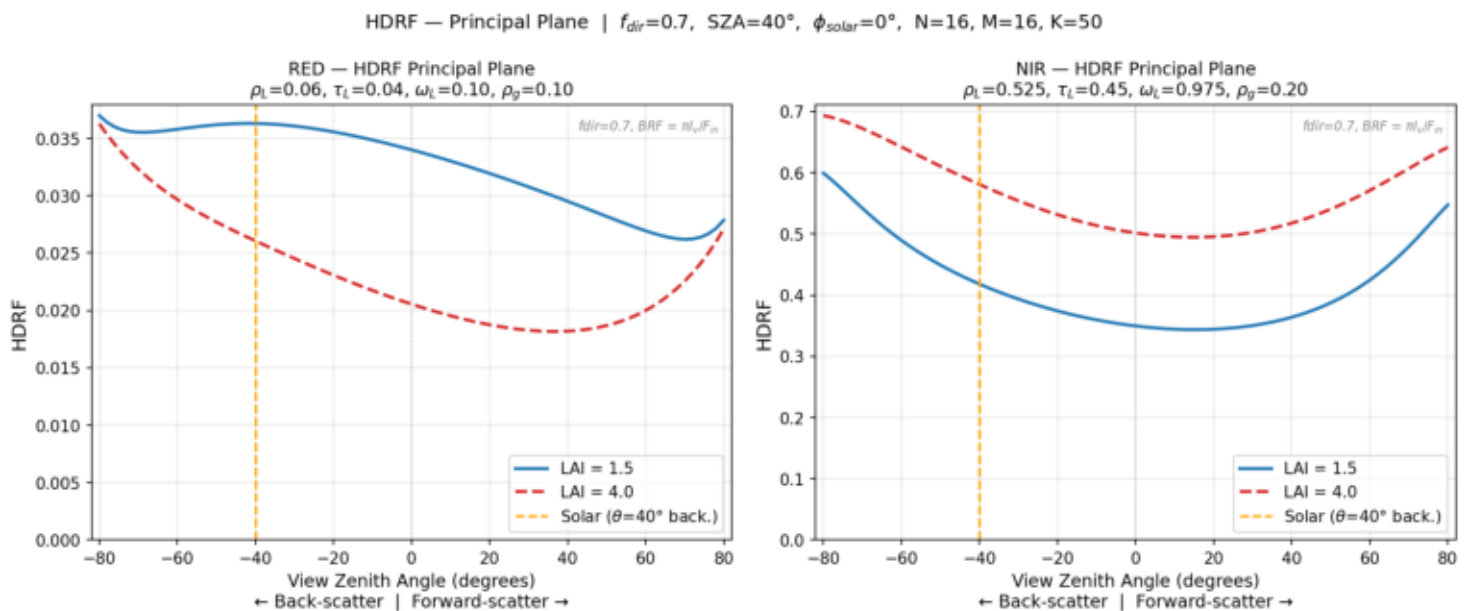
Same three-panel uncollided DOM solution for the NIR band ($\rho_{oL}=0.525$, $\tau_{oL}=0.45$, $\rho_{oG}=0.20$). NIR leaves are nearly transparent ($\omega_{oL}\sim 0.975$), so substantially more radiation penetrates to the ground and the upward flux is much higher than in the RED band. Plot C shows the uncollided BRF is also significantly larger in NIR due to the high leaf single-scattering albedo. Parameters identical to Figure 4.

Figure 6 — HDRF Hemispherical Polar Map (fdir=0.70)



2x2 hemispherical polar maps of HDRF for RED and NIR bands at LAI=1.5 and 4.0. Polar orientation: 0 deg at top (forward scatter direction), 180 deg at bottom (sun sky position). Yellow star marks the sun at SZA=40 deg. With mixed illumination (fdir=0.70), the HDRF pattern is smoother than pure BRF — the diffuse component fills in angular variations. NIR HDRF is 10-20x higher than RED due to high leaf albedo (0.90 vs 0.10). Increasing LAI from 1.5 to 4.0 reduces the soil contribution and flattens the angular pattern. Parameters: SZA=40 deg, $\phi_{solar}=0$ deg, $N=M=16$, $K=50$.

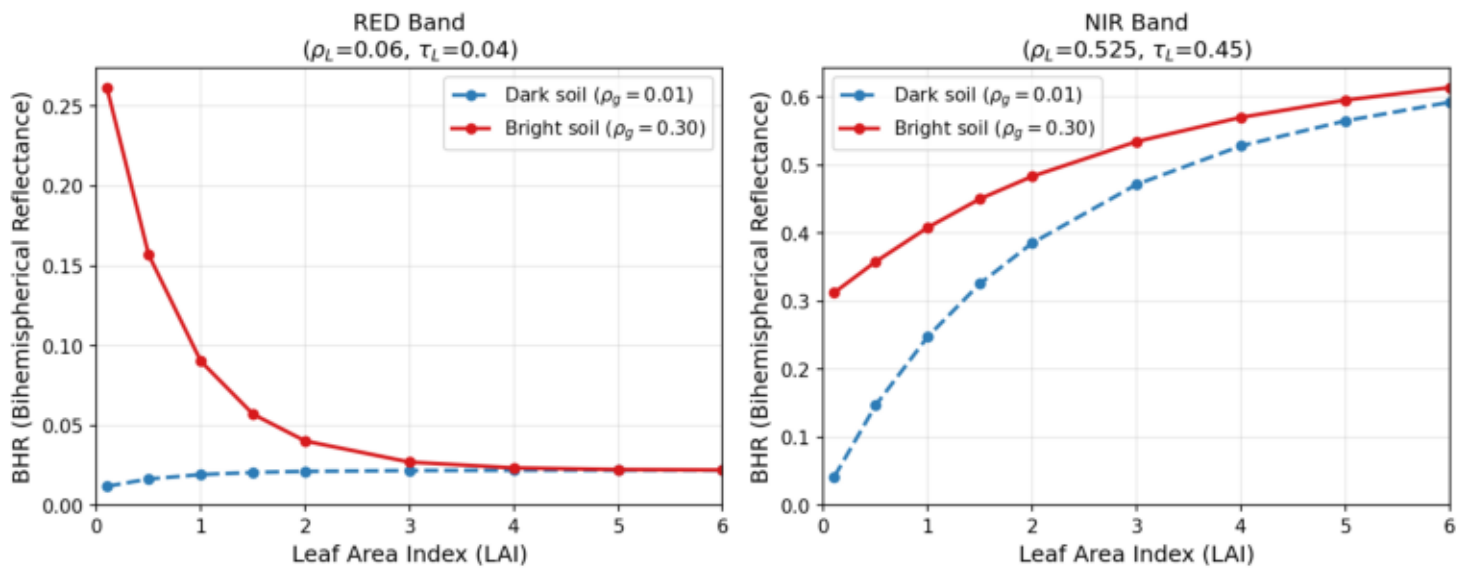
Figure 7 — HDRF Principal Plane Cross-Section (fdir=0.70)



HDRF vs. view zenith angle in the principal plane for RED (left) and NIR (right) bands at LAI=1.5 and 4.0. Negative VZA = backscatter toward sun; positive VZA = forward scatter away from sun. Orange dashed line marks the solar retro-illumination direction at VZA=-40 deg. The smooth bowl shape (minimum near nadir) is a consequence of the bi-Lambertian phase function used in DOM_v3, which has no hot-spot singularity at the retro-illumination direction (Ch4 Sec.9 not implemented). Parameters: SZA=40 deg, fdir=0.70, N=M=16, K=50.

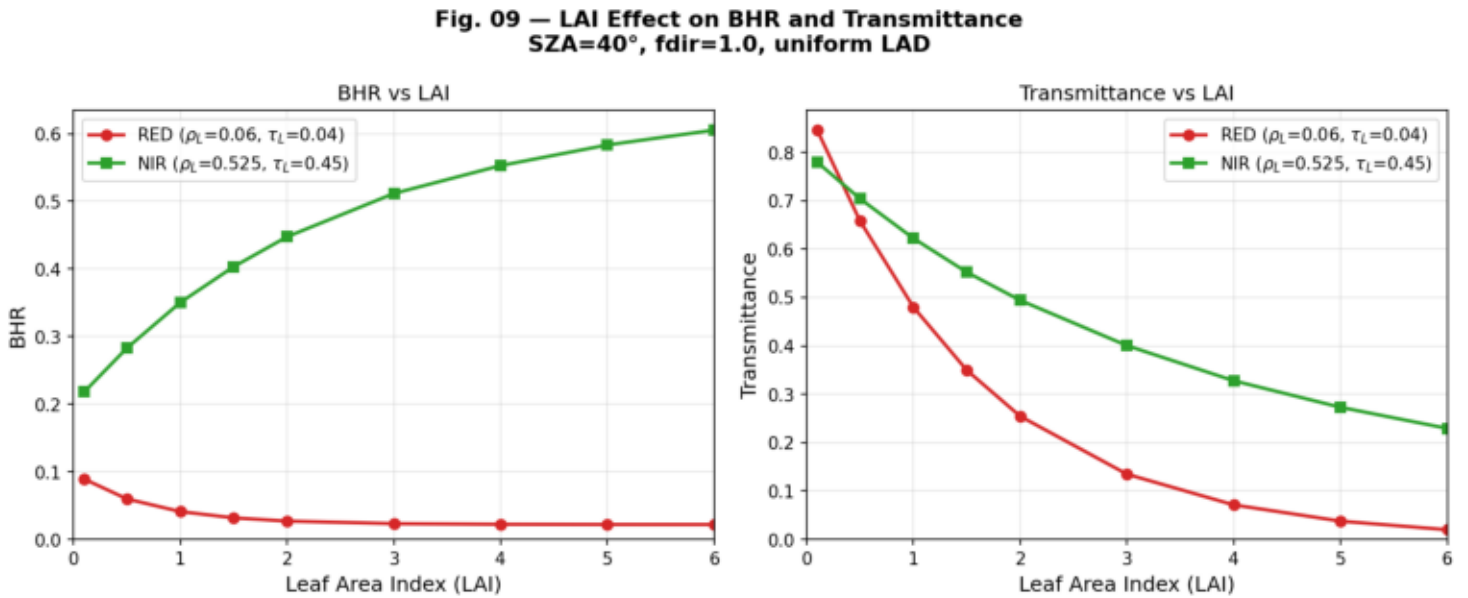
Figure 8 — BHR as a Function of LAI and Soil Brightness

Fig. 08 — BHR as a Function of LAI and Soil Brightness
SAZ=40°, fdir=1.0, uniform LAD



Bihemispherical reflectance (BHR = upward flux at canopy top / F_{in}) vs. LAI for dark ($\rho_g=0.01$) and bright ($\rho_g=0.30$) soil. RED band: BHR decreases monotonically with LAI as absorbing leaves progressively mask the bright soil; both soil cases converge to the same leaf-limited value at high LAI. NIR band: BHR increases with LAI for dark soil (leaf reflectance exceeds soil reflectance) but decreases for bright soil (leaves mask the highly reflective ground). Both cases converge at large LAI to the canopy-only reflectance. Parameters: SAZ=40 deg, fdir=1.0, uniform LAD ($G=0.5$).

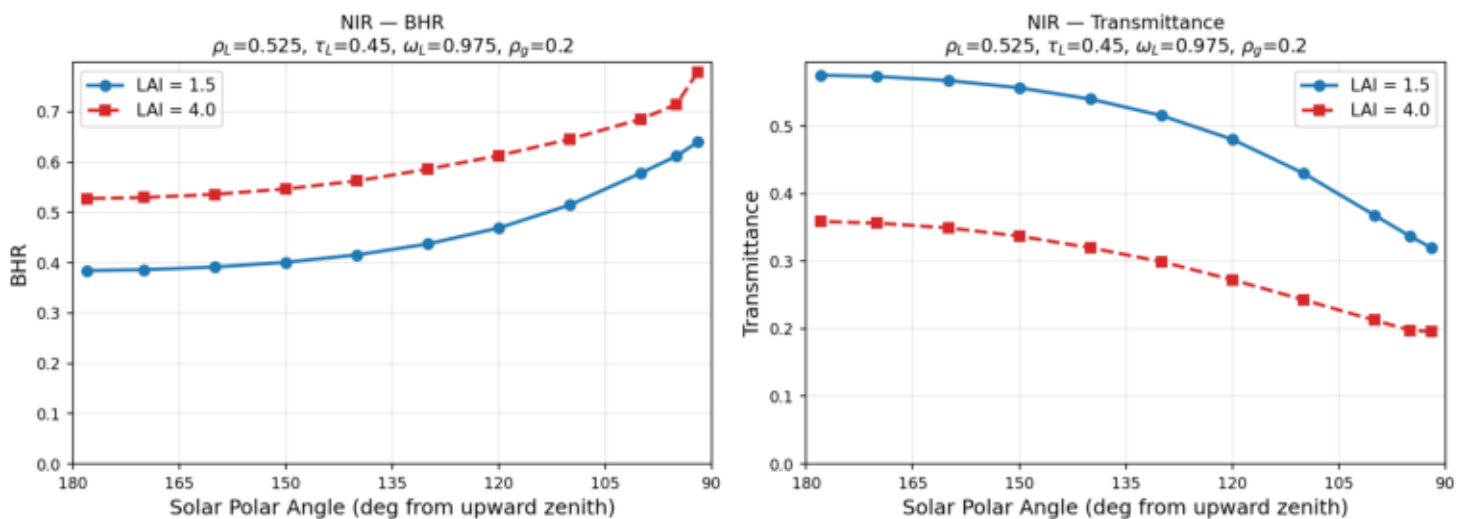
Figure 9 — LAI Effect on BHR and Transmittance



BHR (left) and canopy transmittance (right) vs. LAI for RED and NIR bands. Transmittance = net downward irradiance at the ground: $(F_d - F_u)_{\text{bottom}} / F_{\text{in}}$. NIR BHR increases steeply with LAI (high leaf scattering albedo) while RED BHR stays near-flat (strong absorption limits multiple scattering). Both transmittances decrease exponentially with LAI as the canopy intercepts more radiation. At LAI=6, NIR transmittance (~15%) still exceeds RED transmittance (~2%) because NIR photons scatter through the canopy rather than being absorbed. Parameters: SZA=40 deg, fdir=1.0, rhoG=0.10 (RED), 0.15 (NIR).

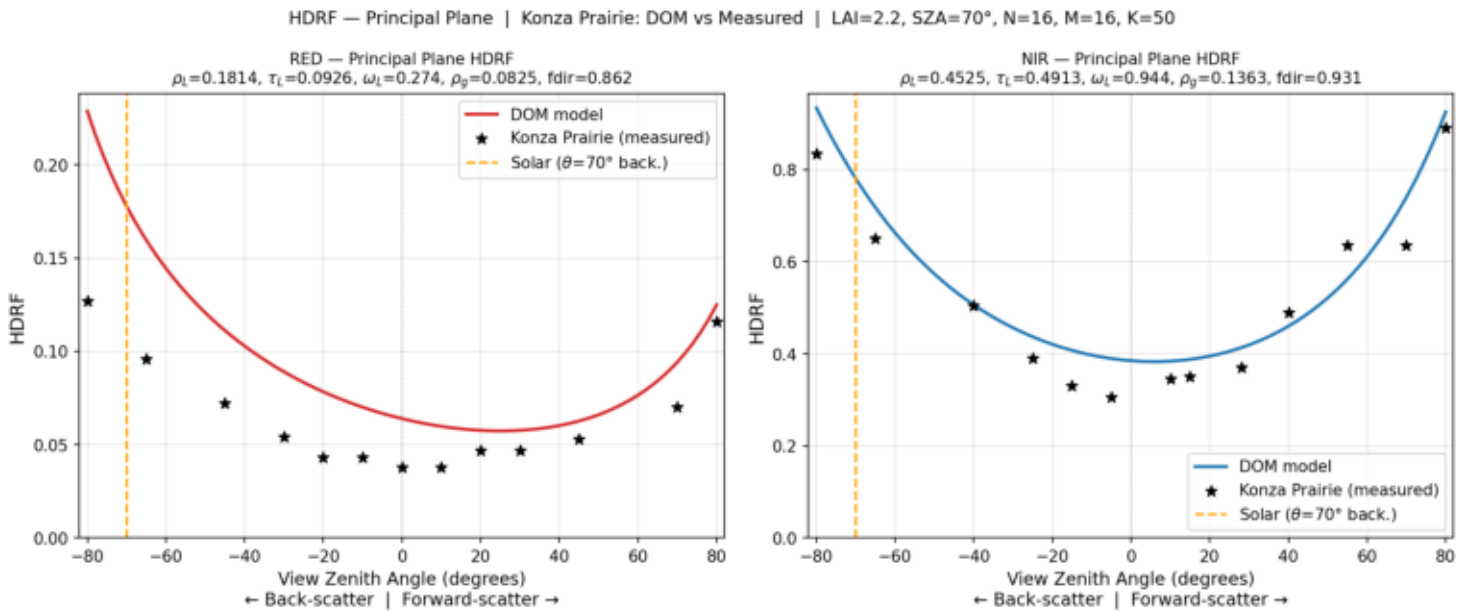
Figure 10 — Solar Polar Angle Effect on BHR and Transmittance

Fig. 10 — Solar Polar Angle Effect on NIR BHR and Transmittance
 $f_{dir}=0.7$, $\rho_g=0.2$, uniform LAD ($G=0.5$)



BHR (left) and canopy transmittance (right) vs. solar zenith angle (SZA) from 10 to 80 degrees at fixed LAI=1.5. NIR BHR increases steeply at high SZA: oblique solar geometry means the beam traverses more leaf area per unit depth, driving more multiple scattering and increasing the upward escape fraction. RED BHR is nearly flat because strong leaf absorption quenches multiple scattering regardless of geometry. Both transmittances decrease with SZA (longer optical path). RED transmittance drops sharply to near zero at SZA=80 deg. Parameters: LAI=1.5, $f_{dir}=1.0$, uniform LAD.

Figure 11 — Konza Prairie: DOM Model vs. Measured HDRF



DOM_v3 principal-plane HDRF (solid line) vs. field measurements from Konza Prairie grassland, Kansas (stars). RED: $\rho_L=0.1814, \tau_L=0.0926, \rho_g=0.0825, fdir=0.862$. NIR: $\rho_L=0.4525, \tau_L=0.4913, \rho_g=0.1363, fdir=0.931$. LAI=2.2, SZA=70 deg. The DOM reproduces the broad bowl-shaped angular dependence but overestimates HDRF at extreme view angles and misses the hot-spot enhancement visible in the field data near VZA=-70 deg (retro-illumination direction). This discrepancy directly demonstrates the need for the hot-spot parameterisation of Ch4 Sec.9, which accounts for shadow-hiding at retro-illumination.